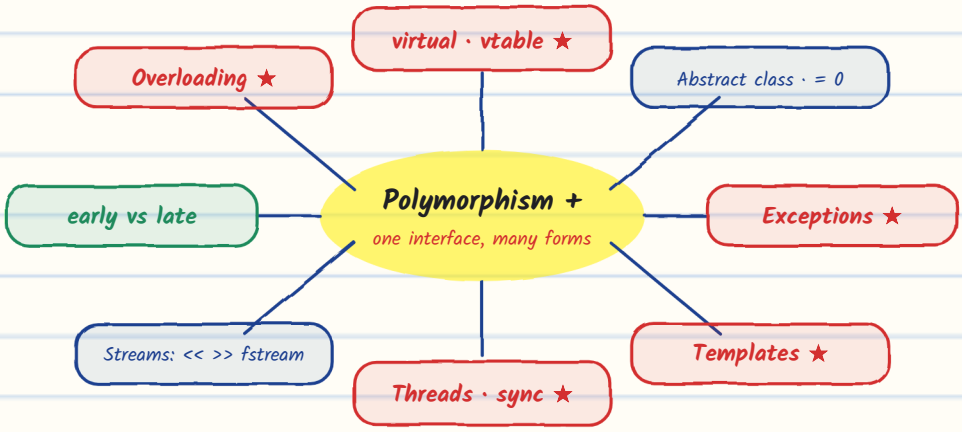
Date: 13.10.2026
topper's notes

Polymorphism, Exceptions, Templates & Threads

Virtual · Exceptions · Templates · I/O · Threads: exam-ready in 11 sheets



↑ the whole lesson on one page: red = high-yield, green = the idea everything else hangs on

➜ Syllabus coverage

The OOP syllabus continues with "Polymorphism: overloading, abstract classes, virtual functions and overriding; Exceptions; Object Oriented I/O; Template functions and classes; Multi-threaded Programming." Classes, access and static members are in the first lesson; constructors and inheritance in the second.

High-yield list

① Overload: compile time; override: run time
② Return type alone ≠ overload
③ No overload: `::` `.` `.*` `?:` `sizeof`
④ Java methods virtual by default
⑤ ≥ 1 pure virtual ⇒ abstract
⑥ Polymorphic base: virtual dtor
⑦ `finally` runs either way
⑧ Checked = compiler-enforced
⑨ Templates: compile-time copies
⑩ `start()` = new thread; `run()` no

Legend yellow = key term · red circle = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

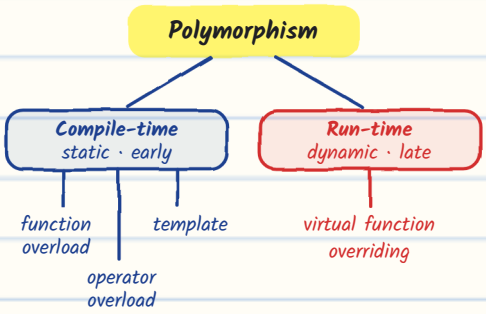
➜ Contents

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1 Polymorphism: Compile-time vs Run-time; Overloading

most tested ★

Definition Polymorphism ("one interface, many forms") lets the same name do different work depending on the types involved. **Binding** = connecting a call to a function body. Early (static) binding happens at compile time — overloading and templates. Late (dynamic) binding happens at run time — overriding through a base pointer or reference.



two families, four tools

Function overloading rules Same name, same scope; parameter lists differ in **number**, **type** or **order**. Return type alone is not enough: `int f(int)` + `double f(int)` = compile error. The compiler ranks: exact match → promotion (`char→int`, `float→double`) → standard conversion → user conversion; two equally good matches = **ambiguity error**. Java also lets a subclass overload an inherited method.

Point	Compile-time polymorphism	Run-time polymorphism
Also called	static, early binding	dynamic, late binding
Decided by	argument list / types	actual object type
Mechanism	overloading, templates	`virtual` + overriding
Needs inheritance	no	yes

```
class Complex {
  double re, im;
public:
  Complex(double r=0, double i=0)
      : re(r), im(i) {}
  Complex operator+(const Complex& c) const {
    return Complex(re + c.re, im + c.im);
  }
};
Complex a(1,2), b(3,4);
Complex c = a + b;  // a.operator+(b) → 4+6i
```

Worked example ✍ which f? `void f(int); void f(double);`
`f('a')` → promotion → f(int)
`f(2.5f)` → float promotes → f(double)
`f(5L)` → long converts to both → **ambiguous**

C++ operators	Operators
Cannot be overloaded	`::` · `.` · `.*` · `?:` · `sizeof` · `typeid` · `alignof` · the `_casts` · `# ##`
Only as a **member**	`=` · `()` · `[]` · `->` (and conversion operators)
Usually a **friend**	`<< >>` for streams; symmetric `a + 2` / `2 + a`

Memory trick The five you can't touch: "**Scope, Dot, Dot-star, Question, Size**" — `:: . .* ?: sizeof`.

Exam trap **Java has no user operator overloading**; `+` on String is built in. `f(int, int = 0)` + `f(int)`: `f(5)` is ambiguous.

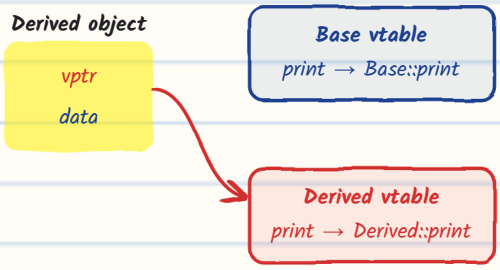
Rules Precedence, associativity and arity never change; no new operators (`**`); one operand must be a user type.

2 Virtual Functions, vtable and Overriding

output questions ★

Definition **Overriding** = a derived class redefines an inherited method with the **same signature**. A C++ **virtual function** is resolved by the **object's** type, not the pointer's, so `Base* p = new Derived; p->f();` runs `Derived::f`. Without `virtual` the pointer's type decides (static binding). The compiler implements this with a **vtable** per class and a hidden **vptr** per object.

```
class Base { public:
  void show() { cout << "Base "; }
  virtual void print() { cout << "BasePrint"; }
};
class Derived : public Base { public:
  void show() { cout << "Derived "; }
  void print() override
              { cout << "DerivedPrint"; }
};
Base* p = new Derived;
p->show();   // Base          (static)
p->print();  // DerivedPrint (dynamic)
// output: Base DerivedPrint
```



p->print() reads vptr, then jumps via vtable

one vtable per class, one vptr per object

Point	Overloading	Overriding
Where	same class	base and derived class
Signature	must **differ**	must be the **same**
Return type	anything	same or covariant (a subclass type)
Binding	compile time	run time
Access (Java)	any	same or wider, never narrower
static / private / final	can be overloaded	**not overridable**; static is hidden

Language	Virtual by default?	Keywords	Block overriding
C++	no	`virtual`, `override`	`final` (C++11)
Java	**yes** (non-static, non-private)	`@Override` (optional)	`final`
C#	no	`virtual` / `override`	`sealed`

Memory trick "**The pointer picks the menu; with `virtual`, the object picks the dish.**"

Exam trap **Constructors cannot be virtual**; a virtual call inside one runs the base version. Java overriding may not **reduce access** or throw broader checked exceptions.

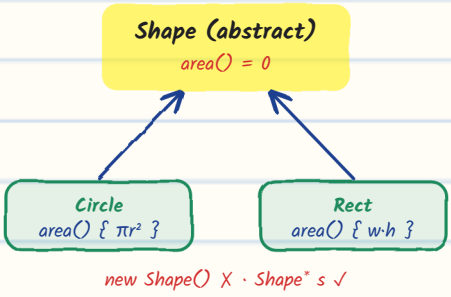
Worked example ✍ sizeof

64-bit: `class A { int x; };` → 4. Add one virtual function → **16** (8-byte vptr + 4 + padding). More virtual functions: still one vptr.

3 Abstract Classes and Virtual Destructors

high-yield ★

Definition A *pure virtual function* is declared with `= 0` — `virtual double area() = 0;` — and says "every subclass must supply this". A class with **at least one** pure virtual function is an *abstract class*: no objects of it can be created, but pointers and references to it can. A derived class that does not override every pure virtual function is abstract too. Java and C# use the keyword `abstract`.



dashed = cannot be instantiated

```
class Shape { public:
  virtual double area() = 0;  // pure
  virtual ~Shape() {}         // virtual dtor
};
class Circle : public Shape { double r;
public:
  Circle(double r) : r(r) {}
  double area() override { return 3.14159*r*r; }
};
Shape* s[2] = { new Circle(1), new Rect(2, 3) };
double total = 0;
for (auto x : s) total += x->area();
// Shape t;  → error: abstract class
```

Can an abstract class …	C++	Java
have constructors, fields, normal methods?	yes (derived class runs the constructor)	yes
have **zero** abstract methods?	no — abstract only if ≥ 1 pure virtual	**yes**, if marked `abstract`
give the abstract method a body?	yes, defined outside: `Shape::area()`	no

Same in Java `abstract class Shape { abstract double area(); }` · `class Circle extends Shape {…}` · `Shape s = new Circle(1);` is legal — the reference type may be abstract.

Virtual destructor `Base* p = new Derived; delete p;` — if `~Base()` is **not** virtual, only the base destructor runs (formally undefined behaviour) and the derived part leaks. Rule: *a class with any virtual function needs a virtual destructor*. Order with it: `~Derived()` then `~Base()`.

Object slicing `Derived d; Base b = d;` copies **only the Base part**; the derived fields are "sliced off" and `b.print()` calls `Base::print`. Polymorphism works only through **pointers or references**, never by value. Java has no slicing: variables hold references.

Memory trick "**`= 0` means zero objects.**" Put `= 0` anywhere in a class and it can no longer be built.

Exam trap "Abstract class contains only abstract methods" is **false**. C++ has no `abstract` keyword. Abstract class vs interface: first lesson.

Worked example ✍ total area

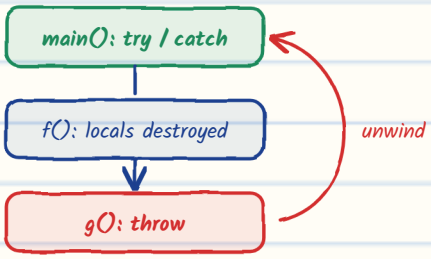
`Circle(1) + Rect(2,3)` = 3.14159 + 6 = **9.14**, each `area()` chosen at run time.

4 Exception Handling: C++ and Java

output questions ★

Definition An **exception** is an abnormal event at run time (divide by zero, bad index, missing file) that breaks the normal flow. `try` guards a block, `throw` raises an exception object, `catch` handles a matching type. A `throw` triggers **stack unwinding**: frames are popped and their locals destroyed until a handler matches; none → `std::terminate()` (C++) or the thread dies with a stack trace (Java).

```
double divide(int a, int b) {
  if (b == 0) throw runtime_error("b = 0");
  return (double)a / b;
}
try {
  cout << divide(10, 0);   // skipped
} catch (const runtime_error& e) {
  cout << e.what();        // b = 0
} catch (...) {            // any type
  cout << "unknown";
}
```



Keyword	C++	Java
`throw`	any type; `throw;` rethrows	`throw new X()` — a Throwable
`throws`	none; `noexcept` = will not throw	header lists checked exceptions
`finally`	**none** — destructors clean up (RAII)	runs **whether or not** one occurs
catch-all	`catch (...)`, must be last	`catch (Exception e)`; multi-catch `A | B`

Java class	Kind	Examples
Throwable → **Error**	unchecked	OutOfMemoryError, StackOverflowError (don't catch)
Exception → **RuntimeException**	unchecked	Arithmetic, NullPointer, ArrayIndexOutOfBounds, ClassCast
other **Exception**s	checked	IOException, FileNotFound, SQLException, InterruptedException

Checked = the compiler forces you to catch it or declare it with `throws`. C++: `std::exception` → *logic_error* (*out_of_range*, *invalid_argument*), *runtime_error* (*overflow_error*, *underflow_error*), *bad_alloc*, *bad_cast*; all have `what()`.

Memory trick "**Catch the child before the parent.**" The first matching handler wins, so derived types go first. "**Checked = Compiler-checked.**"

Exam trap `catch (Exception e)` above a subclass catch = **compile error** in Java. `finally` is skipped only by `System.exit()` or a crash; a `return` in `finally` overrides the try's.

Worked example ✍ **trace**

```
int[] a = new int[3];
try { a[3]=10;
print("X"); } catch
{print("Y");} finally
{print("Z");}
print("W"); →
```

YZW

5 Template Functions and Classes; Java Generics

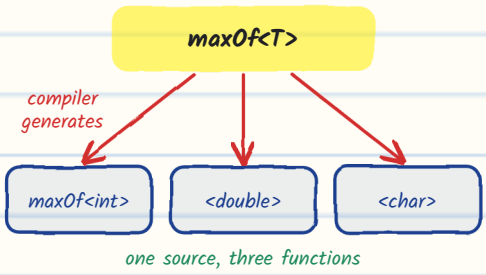
programming question ★

Definition A *template* is a blueprint written once with a type parameter T. When code uses it with a concrete type, the compiler *instantiates* it — generates a separate function or class for that type — at **compile time**. This is **parametric (generic) polymorphism**; the whole STL (vector<T>, sort, map<K,V>) is built on it.

```
template <typename T>
T maxOf(T a, T b) { return a > b ? a : b; }

cout << maxOf(3, 7)        // T = int
     << " " << maxOf(2.5, 1.5)
     << " " << maxOf('a', 'z');
// output: 7 2.5 z
// maxOf(3, 2.5)  → error: T int or double?
// maxOf<double>(3, 2.5) → 3
```

```
template <class T, int N = 100>
class Stack {
  T data[N];  int top = -1;
public:
  void push(const T& x) {
    if (top == N - 1) throw overflow_error("full");
    data[++top] = x;
  }
  T pop() {
    if (top < 0) throw underflow_error("empty");
    return data[top--];
  }
};
Stack<int> si;  Stack<string, 10> ss;
```



instantiation at compile time

Specialization A **full specialization** replaces the template for one exact type: template<> const char* maxOf(const char* a, const char* b) using strcmp — without it two C strings are compared by **address**. Class templates may also be **partially** specialized (Stack<T*>); function templates cannot. Template code lives in the **header**: the compiler needs the full body to instantiate it.

Point	C++ template	Java generics
Resolved	compile time, per type	compile time, then **type erasure**
Code copies	one per type (code bloat)	one class (Object inside)
Primitive args	yes: Stack<int>	no: List<Integer>, never <int>
Extras	non-type params <int N>, specialization	bounds <T extends Comparable<T>>, wildcard ?

Memory trick "**Templates are cookie cutters**": one cutter, a new cookie for every dough (type). Java generics: one cookie with a label.

Exam trap Java generics are **not** templates: after erasure List<String> and List<Integer> share **one class**, so new T() is illegal.

Worked example ✍

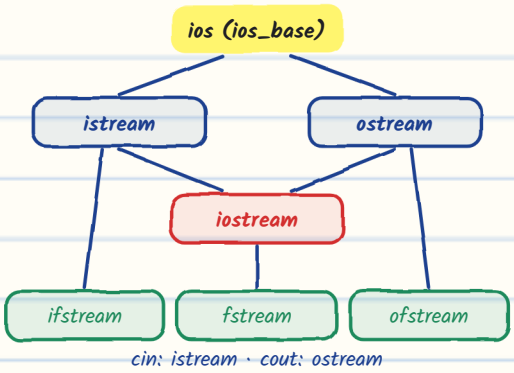
Java

static <T> void show(T v): show(100) autoboxes to Integer. Box<String> b = new Box<>(); — the diamond infers String.

6 Object-Oriented I/O: Streams

MCQ facts

Definition In OO I/O, data moves through **stream objects**: a stream is a sequence of bytes flowing from a source (keyboard, file) or to a sink (screen, file). C++ supplies a class hierarchy in <iostream>, <fstream>, <sstream>; << (**insertion**) and >> (**extraction**) are overloaded operators, so user classes can join in.



cin: istream · cout: ostream

iostream: multiple inheritance

Object	Class	Device	Buffered
cin	istream	keyboard	yes
cout	ostream	screen	yes
cerr	ostream	error	no
clog	ostream	error log	yes

Mode	Meaning
ios::in · out	read · write
ios::app	every write at the end
ios::ate	open, seek to end once
ios::trunc	empty the file first
ios::binary	no text translation

```
class Point { int x, y;
 public:
  Point(int a=0, int b=0) : x(a), y(b) {}
  friend ostream& operator<<(ostream& os,
                             const Point& p) {
    return os << "(" << p.x << ","
              << p.y << ")";
  }
};
ofstream out("pts.txt", ios::app);
out << Point(2, 3) << endl;  // (2,3)
out.close();
ifstream in("pts.txt");  string line;
while (getline(in, line)) cout << line;
```

Manipulator	Effect
endl	newline + **flush**
setw(n)	width, **next** item only
setprecision(n)	digits (after point with fixed)
setfill(c)	padding character
hex oct dec	number base (sticky)

ofstream opens with out | trunc by default; check a file with is_open(), eof(), fail().

Java I/O **Byte** streams: InputStream / OutputStream (FileInputStream). **Character** streams: Reader / Writer (FileReader, FileWriter, BufferedReader with readLine()). System.in is an InputStream, System.out a PrintStream; Scanner parses tokens (nextInt(), nextLine()). Close files with **try-with-resources**; save whole objects with ObjectOutputStream (class must be Serializable).

Memory trick "**Arrows point where the data goes**": cout << x pushes x out; cin >> x pulls into x.

Exam trap operator<< must be a **non-member** (friend) returning ostream& for chaining. cin >> s stops at a space.

Worked example cout << setw(5) << 42 << 7; → " 427": setw pads only 42.

7 Threads: Creation and Life Cycle high-yield ★

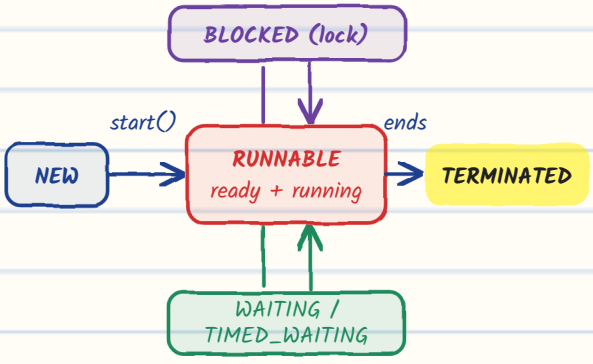
Definition A thread is the smallest unit of CPU scheduling — a "lightweight process". Multi-threading runs two or more threads inside one process; they share its code, data and heap, but each has its own **stack, registers and program counter**. One core: threads run **concurrently** (interleaved); many cores: in **parallel**. Uses: responsive GUIs, servers, background downloads.

Point	Process	Thread
Memory	own address space	shares the process's memory
Create / switch	heavy, slow	light, fast
Communication	IPC: pipes, sockets	shared variables (need locks)
One crashes	others unaffected	can kill the whole process

```
// Way 1: extend Thread
class MyThread extends Thread {
  public void run() { System.out.println("T1"); }
}
// Way 2: implement Runnable (preferred)
class Task implements Runnable {
  public void run() { System.out.println("T2"); }
}
new MyThread().start();
new Thread(new Task()).start();
new Thread(() -> System.out.println("T3"))
    .start();              // lambda (Java 8)
```

start() vs run() `start()` asks the JVM for a **new** thread, which then calls `run()`. Calling `run()` directly is an ordinary call on the **current** thread. `start()` twice → IllegalThreadStateException. **Runnable is preferred**: the class can still extend another class (single inheritance), and one task can be shared by many threads. The order in which threads print is **not guaranteed**.

Method	Does
`sleep(ms)`	pause; **keeps** locks
`join()`	wait for it to end
`yield()`	hint: let others run
`wait()`	**releases** the lock
`setPriority`	1 … 5 … 10
`setDaemon`	background thread



Java Thread.State: six states

Memory trick "**N**ew **R**unners **B**lock, **W**ait, **T**ime-wait, **T**erminate" — the six Java states in enum order.

Exam trap Java has **no RUNNING state**: RUNNABLE = ready + running. Waiting for a lock = BLOCKED; `wait()`/`join()` = WAITING; `sleep(ms)` = TIMED_WAITING.

Worked example ✍ `t.run(); print("M");` → run's output always first (same thread). `t.start(); print("M");` → order **unpredictable**

8 Synchronization, Race Conditions and Deadlock

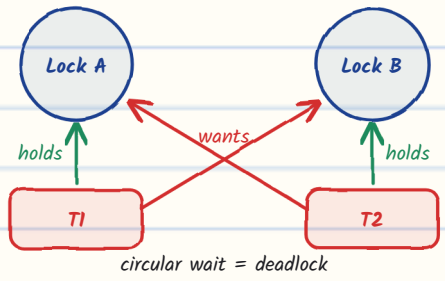
written favourite ★

Definition A **race condition** occurs when two threads access shared data, at least one writes, and the result depends on the timing. The code that touches shared data is the **critical section**; **synchronization** guarantees **mutual exclusion** — only one thread in it at a time. `count++` looks like one step but is three (read, add, write), so two threads can lose an update.

```
// Java: monitor lock on 'this'
class Counter {
  private int count = 0;
  public synchronized void increment() {
    count++;
  }
  public int get() { return count; }
}
// or: synchronized (lock) { count++; }
// or: AtomicInteger c; c.incrementAndGet();
```

```
// C++11: <thread>, <mutex>
int count = 0;  mutex m;
void work() {
  for (int i = 0; i < 1000; i++) {
    lock_guard<mutex> g(m);  // RAII lock
    count++;
  }
}
thread t1(work), t2(work);
t1.join();  t2.join();   // wait for both
cout << count;           // always 2000
```

Tool	What it gives
Mutex (lock)	one owner at a time; the locker must unlock
Semaphore	counter: up to *n* threads; binary semaphore ≈ mutex (but no ownership)
Monitor	lock + waiting in one object: `synchronized` + `wait/notify`
`volatile` (Java)	visibility of the latest value; **not atomicity** — `count++` still races
`join` / `detach` (C++)	wait for it / let it run alone; neither → `std::terminate`



each holds one, waits for the other

Deadlock: 4 Coffman conditions ① **Mutual exclusion** — a resource is held by one thread. ② **Hold and wait** — hold one, wait for another. ③ **No preemption** — a lock cannot be taken away. ④ **Circular wait** — T1 → T2 → … → T1. All four are needed; break one and deadlock is impossible — usually a **fixed lock order** (always A before B). **Starvation** = never gets a turn; **livelock** = busy but no progress.

Memory trick Coffman = "**My Hat, No Coat**": Mutual exclusion, Hold & wait, No preemption, Circular wait.

Exam trap `wait()/notify()` are **Object** methods, called only while holding the lock (else `IllegalMonitorStateException`).

Worked example ✍ lost update

count = 5. T1 reads 5, T2 reads 5, T1 writes 6, T2 writes 6 → final **6**, not 7. Without the lock, two × 1000 increments give ≤ 2000.

Night-Before-the-Exam Sheet

Fact sheet

Overloading → compile time, early binding
Return type alone cannot overload
Member only: = () [] ->
vtable per class, vptr per object

`virtual f() = 0` → abstract class

C++ has no finally (RAII); Java finally always (except System.exit)
Unchecked: RuntimeException + Error

Java generics: type erasure, no primitives
Thread shares heap; own stack + PC

Overriding + virtual → run time, late binding
No overload: `::` `.` `.*` `?:` `sizeof`
Java: no user operator overloading
Constructor never virtual; destructor should be

Polymorphism via pointer/reference; by value = slicing
Checked: IOException, SQLException

Templates: instantiated per type at compile time

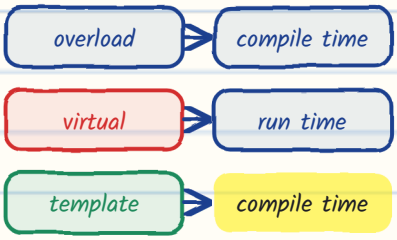
`endl` = newline + flush; `cerr` unbuffered
Java: 6 thread states; priority 1–10, default 5

→ Top 12 one-liners for MCQs

1. Polymorphism = one interface, many forms
2. Function overloading = static polymorphism
3. Java methods are virtual by default
4. C# needs `virtual` + `override`; `sealed` stops it
5. Catch derived exceptions before base
6. `catch (...)` catches anything, goes last
7. `throw` raises; `throws` declares
8. `noexcept` violated → `std::terminate`
9. STL is built on templates
10. `<<` overload = friend, returns `ostream&`
11. `start()` new thread; `run()` same thread
12. Deadlock needs all four Coffman conditions

Top traps return type ≠ overload · virtual call in a constructor runs the base version · abstract class may have constructors · Java has no RUNNING state · `volatile` ≠ atomic · `sleep()` keeps the lock, `wait()` releases it · catch(Exception) before a subclass = compile error.

All tricks together
Scope, Dot, Dot-star, Question, Size · pointer picks the menu, object picks the dish · = 0 means zero objects · catch the child before the parent · cookie cutters · arrows point where data goes · New Runners Block... · My Hat, No Coat.



Worked example ✍ one keyword changes the output In the Base/Derived program, remove `virtual` from `print()`: `p->print()` is now bound to the pointer type, so the output becomes "Base BasePrint" instead of "Base DerivedPrint".

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. Function overloading is resolved at:
   (a) compile time (b) run time
   (c) link time (d) load time

2. Which operator cannot be overloaded in C++?
   (a) + (b) []
   (c) == (d) ::

3. A C++ class with at least one pure virtual function is:
   (a) a friend class (b) a template class
   (c) an abstract class (d) a static class

4. In Java, which block runs whether or not an exception occurs?
   (a) catch (b) finally
   (c) throw (d) throws

5. To begin a new thread in Java, call:
   (a) `run()` (b) `init()`
   (c) `start()` (d) `execute()`

6. Which is an unchecked exception in Java?
   (a) IOException (b) SQLException
   (c) EOFException (d) ArithmeticException

7. Why make a base-class destructor virtual?
   (a) so `delete` via `Base*` runs `~Derived` (b) to make the class abstract
   (c) to block copying (d) to speed up `delete`

8. C++ templates are instantiated at:
   (a) run time (b) compile time
   (c) load time (d) link time only

9. How many states does Java's `Thread.State` define?
   (a) 6 (b) 4
   (c) 5 (d) 7

10. Which is **not** a necessary condition for deadlock?
    (a) mutual exclusion (b) hold and wait
    (c) circular wait (d) preemption

Answer key → 1-a · 2-d · 3-c · 4-b · 5-c · 6-d · 7-a · 8-b · 9-a · 10-d

Written question Distinguish compile-time and run-time polymorphism. Explain how a virtual function gives run-time polymorphism in C++, with an example and the vtable mechanism. Why should the destructor of a polymorphic base class be virtual? 10 marks

1. **Distinction (3)** — compile-time: overloading / templates, chosen from the argument list, early binding, fast; run-time: overriding of virtual functions through a base pointer or reference, chosen by the object's type, late binding, needs inheritance. 3
2. **Example (3)** — `Base` with `virtual void print()`, `Derived` overriding it; `Base* p = new Derived; p->print();` prints the derived version; without `virtual`, the base version. 3
3. **Mechanism (2)** — each class with virtual functions has a vtable of function addresses; each object has a hidden vptr to its class's vtable; the call reads vptr, indexes the table and jumps. 2
4. **Virtual destructor (2)** — `delete p` with a non-virtual `~Base` runs only the base destructor (undefined behaviour, leak); a virtual one runs `~Derived` then `~Base`. 2

→ Before the exam ✓

- ✓ Can list the non-overloadable operators
- ✓ Can trace a virtual / non-virtual call
- ✓ Can write an abstract class in C++ and Java
- ✓ Can order catch blocks and trace finally
- ✓ Can write a function and class template
- ✓ Can overload << for a class
- ✓ Can draw the six Java thread states
- ✓ Can state the four deadlock conditions